

Ariful Islam

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[Ariful Islam](#)

Education

Master of Science in Mechatronics Engineering

CGPA: **4.00**/4.00

Rajshahi University of Engineering and Technology (RUET), Bangladesh

April 2025-Present

Relevant Courses: Advanced Control Systems, Advanced Robotics, Avionics for Mechatronics Engineers, Advanced Electromechanical Drives, Advanced Mechatronics System, and Advanced Mechatronics System Design.

Thesis Title: Non-destructive Fruit Ripeness Detection: Sensor Fusion for Intelligent Harvesting Decision

Bachelor of Science in Mechatronics Engineering

CGPA: 3.44 (**3.80** in last 100 credits)/4.00

Rajshahi University of Engineering and Technology (RUET), Bangladesh

Graduation: October 2023

Relevant Courses: Control Systems, Robotics, Artificial Intelligence, Automation, Microcontroller and Interfacing, Digital Signal Processing and Machine Vision, Pneumatic and Hydraulic Control, Power Electronics, Machine Learning Algorithms, Sensors and Instrumentation, Embedded Systems.

Thesis Title: Physician's Prescribed Handwriting Recognition Using Image Processing and Deep Learning Algorithms

Experience

Assistant Engineer

Operations Department, Chandpur **115MW** Power Generations Limited, Bangladesh

February 2024-Present

- Operating and monitoring 4x 18V51/60TS and 2x 20V45/60 MAN **IC engines**.
- Developing and implementing Fuel and Water Treatment Systems, Compressed and Charge Air System, Cooling System, Exhaust Gas Boiler, Lube Oil Treatment, and Firefighting System.
- Installation, monitoring, and operation of 11KV/400V Transformer, and 11/132KV Substation.
- Validating electric components (sensors, PLCs, limiters, switches, solenoids, etc.)
- UPS, DC System, SaCoS and PowerOS **SCADA** operation.

Graduate Research Student

Dept. of Mechatronics Engineering, Rajshahi University of Engineering and Technology

April 2025- Present

Supervisor: Dr. Zinat Tasneem, Associate Professor

- Sensor-driven fruit maturity analysis.
- Sensor-fusion technology implementation in agricultural harvesting.
- Real-time embedded device deployment with TinyML

Undergraduate Research Student

Dept. of Mechatronics Engineering, Rajshahi University of Engineering and Technology

March 2022- October 2023

Supervisor: Md Manirul Islam, Assistant Professor

- Pedestrian **Trajectory Prediction** (PTP) based artificial intelligence model analysis.
- Custom **Transfer Learning** Driven Deep-CNN model implementation for machine vision tasks.
- In-depth study on research methodology (**Systematic Review**, Scoping Review, **PRISMA**, Meta-Analysis, etc.) and tools (LaTeX, Mendeley, etc.).

Publications

- "State-of-the-art review on fall prediction among older Adults: Exploring edge devices as a promising approach for the future."; DOI: [10.1016/j.measen.2025.101878](https://doi.org/10.1016/j.measen.2025.101878).
- "Towards Smart Healthcare: Transfer Learning Approach to Detect Offline Handwritten Medicine Names in Real-Time." DOI: [10.1109/ICCIT68739.2025.11491625](https://doi.org/10.1109/ICCIT68739.2025.11491625)
- "Quantifying Methane Emissions and Energy Recovery Potential from Landfill Sites: Insights from Statistical Machine Learning and Predictive Models". DOI: [10.1002/tqem.70179](https://doi.org/10.1002/tqem.70179)
- "Pedestrian Safety through Trajectory Prediction: State-of-the-Art Technologies, Strategies, Challenges, and Future Directions- A Scoping Review." (Submitted). Authors: Ariful Islam et al.
- "Identification of an Optimal Post-Injection Timing-Quantity Window for Simultaneous Soot Reduction and Performance Preservation in a HSDI Diesel Engine." (Under Review). Authors: Sk Tanzim, Dr. Kazi Mostafijur Rahman, Ariful Islam, Ahabab Faiyaz Haque.

Projects

- **Implementation of Free Governor Mode of Operation (FGMO) in an HFO-fired power generation plant for frequency regulation in Bangladesh National Power Grid.**
⇒ Involves configuring and implementing electronic governor control logic using FGMO (Free Governor Mode of Operation) parameters to adjust generator power output based on grid frequency deviations automatically.
- **Optimizing Large Language Model (LLM) output using the Retrieval Augmented Generation (RAG) technique.**
⇒ Processed power plant technical documentation using OCR and optimized with RAG and LLM models to ensure seamless plant operations.
- **Design and construction of a low-cost, handmade DC motor with speed-controlling capability.**
⇒ Construction of a DC motor with the cheapest recycled materials from waste components and fine-tuned design of a PID controller capable of controlling the motor's speed accordingly.
- **IoT-based automatic attendance system deployment in the classrooms of the Mechatronics Engineering department.**
⇒ NodeMCU (ESP32) equipped IoT device construction for detecting student ID with a sensor and fed to a server for keeping the logs.

Skills

- **Programming Language:** Python, C/C++, R.
- **Design and Simulation:** NVIDIA Omniverse, Isaac Sim, Isaac Lab, AutoCAD, SolidWorks, MATLAB, Proteus, LOGO!-Siemens, ROS, Gazebo.
- **Tools:** Tensorflow, PyTorch, Keras, Scikit-Learn, OpenCV, Numpy, Pandas, Matplotlib, SCADA.
- **Productivity Software:** MS Office, LaTeX.

Standardized Test Scores

- **GRE:** 300, **AWA:** 3.5
- **IELTS:** 7.0 (L:7.5, R: 7.0, W: 6.5, S: 6.5)

Trainings & Certifications

- Post Graduate Diploma (PGD) in Logistics and Supply Chain Management, EduPro-UK *February, 2026*
- Industrial Training, Engineering Department, General Facility, Renata PLC *April, 2022*
- Internet of Things (IOT), Bangabandhu Sheikh Mujib Hi-Tech Park – **Result:** 77% *July, 2022- September 2022*
- Workshop on Programmable Logic Controller (PLC), RUET *February, 2018*

Extra-Curricular Activities

- Assistant Technical Secretary, Robotic Society of RUET *2022-2023*
- Event Management Secretary, Onuronon, RUET Cultural Club *2022-2023*

Awards

- University Technical Education Scholarship, RUET *2019-2023*
- Primary Education Scholarship, Dhaka Board *2010*

References

Dr. Zinat Tasneem

Associate Professor

Department of Mechatronics Engineering, Rajshahi University of Engineering & Technology

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Dr. Dip Kumar Saha

Assistant Professor

Department of Mechatronics Engineering, Rajshahi University of Engineering & Technology

Email: dip07me@mte.ruet.ac.bd

Md Manirul Islam

Assistant Professor

Department of Mechatronics Engineering, Rajshahi University of Engineering & Technology

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